

HW 2
To be completed by Monday 2/16

1. This exercise will give you practice coding and manipulating discrete choice models in the simplest setting - a binary choice problem. Here we will examine the determinants of attending a 4-yr college relative to a 2-yr college among post-secondary attendees. Use “dataHW2 Problem1.mat”. The data contains an outcome variable Y which is an indicator for attending a 4-yr college. X_i are student observable characteristics including a constant (these coefficients must vary across choices). The second column of X_i is an indicator for whether either parent graduated from a 4-yr college, and the third column of X_i is the student’s high school GPA. Z contains choice specific attributes (the coefficients are common across choices). The first column of Z captures the distance to the closest 2-year college in miles. The second column of Z captures the distance to the closest 4-year college.
 - (a) Estimate a logit model of the 4-yr college choice using X_i and Z as regressors. Report coefficient estimates and SEs.
 - (b) Calculate the average marginal effect of having a parent with a 4-yr college degree. Calculate the average marginal effect of having a parent with a 4-yr college degree for those students who have a parent with a 4-yr college degree. Provide some intuition for why the latter marginal effect is smaller.
 - (c) Use the delta method to calculate the SE of the overall average marginal effect of having a parent with a 4-yr college degree.
 - (d) Use a non-parametric bootstrap method to calculate the SE of the overall average marginal effect of having a parent with a 4-yr college degree. Resample from the data 500 times and re-estimate the model and associated marginal effect. Report the standard deviation of the estimated marginal effects. The ‘`datasample()`’ command can be useful here.
 - (e) Estimate a probit model of the 4-yr college choice using X_i and Z as regressors. Report coefficient estimates and SEs.
 - (f) Calculate the average marginal effect of having a parent with a 4-yr college degree using the probit results. Is it similar to the logit marginal effect?
 - (g) Use ‘`statacheck.do`’ along with ‘`HW2 P1.csv`’ to check your answers to parts (a)-(f).
 - (h) Just a quick sanity check....would you want to give the logit or probit coefficients (or MFXs) a causal interpretation? If not, which assumption is likely incorrect?

2. Use “dataHW2 Problem2.mat” to complete this exercise. In this problem you will work with a multinomial choice model. Upon graduating high school, individuals have four options: work ($Y=0$), attend a 2-year college ($Y=1$), attend a 4-year public college ($Y=2$), or attend a 4-year private college ($Y=3$). The available student demographic characteristics (and choice specific constants) are as in the previous example. The three columns of X_i are the constant, an indicator for whether either parent graduated from a 4-year college, and the student’s high school GPA respectively. There are two choice specific variables, distance (Z_{dist}) and price (Z_{price}) measured in thousands of dollars. Each column of these matrices reflects the choice specific attribute for the corresponding choice. In other words, the first column are the attributes for the work choice, the second column for the 2-year choice, and so on. Note that I assume that distance and price for the work option are zero.
- Estimate a multinomial logit model with X_i and Z_{dist} and Z_{price} as regressors *using the default `fminunc` options*. Normalize the utility of work to zero. Report parameter estimates and SEs.
 - Re-estimate the model from part (a) (again using the default `fminunc` options), but use different starting values (all zeros/random draws). Report parameter estimates and SEs. Do you get the same results? If not, what do you think is happening?
 - Use the “`optimoptions`” command to set the options such that the estimation results using either the starting values from (a) and (b) are identical.
 - Find the average marginal effect of having a parent with a 4-yr college degree on the likelihood of attending **any** 4-year college.
 - Find the average predicted probability for each choice in the sample. These numbers should match the empirical probabilities observed in the data (check this). What parameters in the model ensure this is the case?
 - Now, assume that the 4-year private option is no longer available. Find the predicted probabilities of the first three choices when the 4th choice is eliminated. Note you do not have to re-estimate the model to this. Find the **percent change** in the predicted choice probability for $Y=\{0,1,2\}$ (relative to when 4-year private is included) and average this across the sample. These numbers should be the same for $Y=\{0,1,2\}$. What feature does this reflect? Do these substitution patterns make sense in the context of the current problem?
 - Find the average change in expected consumer surplus (in thousands of dollars) when the private 4-year option is eliminated. Find the average change in expected consumer surplus separately for individuals whose parents do and do not have a college degree. Note that this is a case where utility is likely not quasi-linear in income, but we’ll do the exercise for practice.